

SyriaTel Customer Churn Analysis

Predicting Customer Churn to Reduce Revenue Loss

Overview

- SyriaTel is losing money every time a customer leaves
- This analysis predicts which customers are likely to churn
- Early identification allows the retention team to intervene

The Business Problem

- Customer churn = lost revenue
- Key question: Are there predictable patterns in customer behavior?

About the Data

- There are 3,333 customer records
- 21 features including call usage, charges and plans
- Only 14.5% of customers churned – imbalanced dataset

Our Approach

- Built 3 models iteratively
- Model 1: Logistic Regression(baseline)
- Model 2: Decision Tree
- Model 3 : Tuned Decision Tree (final model)

How Well Does It Work?

- Our model correctly identifies 94% of cases
- When it flags a churner, it is right 88% of the time
- Model performs consistently on both training and test data

Recommendations

- Deploy the model to flag at-risk customers automatically
- Focus retention efforts on high day charge customers
- Offer discounts or upgrades to flagged customers

Next Steps

- Try Random Forest for better performance
- Address class imbalance using SMOTE
- Monitor model performance over time

Thank you